

Water Resources

Social Science — Geography • Chapter 3 • Chapter Summary & Study Guide for Daksh

Three-fourths of the Earth is covered with water — yet over a billion people face water scarcity. Why? Because only 2.5% of all water is fresh, and only a tiny fraction of that is easily accessible. India receives plenty of monsoon rain, but its distribution in time and space is very uneven. This chapter looks at India's water resources, the case of multipurpose projects (and their critics), and traditional and modern methods of rainwater harvesting.

Why do we have water scarcity?

Water scarcity is not just about quantity — it is also about quality, distribution and access. Causes:

- **Rapid population growth** increases demand for drinking, food and industry.
- **Over-exploitation of groundwater** for irrigation (esp. in Punjab, Haryana, parts of UP).
- **Industrial and urban demand** rises with economic growth — cities draw groundwater faster than recharge.
- **Pollution** from sewage, industrial effluents, fertiliser run-off makes existing water unusable.
- **Unequal access** — different social groups have unequal access to clean water.

Multipurpose river projects — boon or bane?

A multipurpose river project uses a single dam to serve many purposes: irrigation, electricity generation, flood control, water supply, navigation, recreation, fish-breeding. Pandit Nehru called dams the 'temples of modern India'. Important examples: **Bhakra-Nangal** (Sutlej), **Hirakud** (Mahanadi), **Damodar Valley Corporation** (Damodar), **Sardar Sarovar** (Narmada).

But multipurpose projects have come under sharp criticism:

- **Sediment** traps behind dams — riverbed becomes shallow downstream.
- **Aquatic life** disrupted — migratory fish can't move upstream.
- **Floods** sometimes *worsen* when dam waters are released suddenly.
- **Displacement** of local communities — Sardar Sarovar displaced thousands; **Narmada Bachao Andolan** emerged in protest.
- **Soil salinisation** due to over-irrigation in Punjab/Haryana.
- Inter-state water disputes (e.g. Krishna-Godavari).

Rainwater harvesting — a sustainable answer

Rainwater harvesting means catching and storing rainwater locally for direct use or to recharge groundwater. India has a tremendous tradition of doing so:

- **Western Himalayas** — '*kuls*' or '*guls*': diversion channels carrying water from glacier streams to fields.
- **Rajasthan** — '*tankas*' (underground tanks) for drinking water at every house in Bikaner, Phalodi, Barmer; '*khadins*' and '*johads*' for agriculture.
- **Western India** — '*roof-top*' rainwater harvesting (esp. in Gendathur village in Karnataka — model village).
- **Bengal** — flooded inundation channels in the floodplains for irrigating paddy.
- **Meghalaya** — '*bamboo drip irrigation*' (200-year-old system using bamboo pipes for drip irrigation of betel/black pepper).

TANKA (Rajasthan)

Underground room-sized tank, catches rooftop rainwater via channel; drinking water for the family for the year.

ROOF-TOP HARVESTING (urban modern)

Roof -> Pipe -> Filter -> Storage tank
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Recharge well -> Aquifer

Modern legal/institutional steps

- Tamil Nadu was the first state to make rooftop rainwater harvesting compulsory for all houses (2003).
- **Watershed management:** an integrated approach to managing surface and ground water resources of a small drainage basin.
- Public awareness campaigns and water conservation policies.

★ MUST-REMEMBER POINTS

- Only 2.5% of Earth's water is fresh; not all is accessible.
- Causes of scarcity: population growth, over-extraction, pollution, unequal access.
- Multipurpose projects: Bhakra-Nangal (Sutlej), Hirakud (Mahanadi), DVC (Damodar), Sardar Sarovar (Narmada).
- Narmada Bachao Andolan: protest against displacement by Sardar Sarovar dam.
- Traditional water harvesting: kuls/guls (Himalaya), tankas/khadins/johads (Rajasthan), bamboo drip (Meghalaya).
- Tamil Nadu first state to mandate rooftop rainwater harvesting (2003).

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